

Question	Marking Guidance	Mark	Comments
02.1	1. Method to ensure all cut surfaces of the eight cubes are exposed to the sucrose solution; 2. Method of controlling temperature; 3. Method of drying cubes before measuring; 4. Measure mass of cubes at stated time intervals;	3 max	Credit valid method descriptions to fulfil mp1, 2 and 3 (no explanation is required). 2. Accept 'at room temperature' for method 4. Accept time intervals between every 5 minutes with maximum of every 40 minutes. 4. Accept 'weigh the cubes at stated time intervals'
02.2	Yes or No (no mark) Calculation of rate per mm^2 for both sets of data, accept answers in the range 1.6×10^{-5} to 1.8×10^{-5} and 1.5×10^{-5} to 1.6×10^{-5};;; Both correct = 3 One correct = 2 Neither correct – look below for max 2 Allow 1 mark for calculation of surface area of two (sets of) cubes 7350 (mm^2) and 14700 (mm^2) Allow 1 mark for calculation of both rates of osmosis shown in first 40 minutes – between 0.12 and 0.13 and between 0.22 and 0.23 If surface area and/or rate of osmosis is incorrect then, allow 1 mark for (their) calculated rate divided by (their) calculated surface area	3	Accept answers not given in standard form or to any number of significant figures $\geq 2\text{sf}$ as long as rounding correct.